1. $\int (x^2+1)(x-2)dx$ $\quad (x^2+1)(x-2) = x^3 - 2x^2 + x - 2$

$\int (x^3 - 2x^2 + x - 2)dx = \int x^3 dx - \int 2x^2 dx + \int x dx + \int x dx - \int 2 dx = \int x^n dx = \frac{x^{n+1}}{n+1}$

$\int x^3 dx = \frac{x^4}{4} + C_1$

$\int 2x^2 dx = 2\int x^2 dx = 2\frac{x^3}{3} + C_2 = \frac{2x^3}{3} + C_2$

$\int x\, dx = \frac{x^2}{2} + C_3$ $\quad \frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} - 2x + C$ $\quad (a \cdot b)^x = a^x \cdot b^x$

$\int 2 dx = 2x + C_4$

$2^x \cdot e^x = (2e)^x$

2. $\int (2e)^x dx = \frac{(2e)^x}{\ln(2e)} + C$ $\quad \int a^x dx = \frac{a^x}{\ln a} + C$

$\ln(2e) = \ln 2 + \ln e = \ln 2 + 1$ $\quad \ln(a \cdot e) = \ln(a) + \ln(e)$

$\frac{(2e)^x}{1 + \ln(2)} + C$

3. $\int \frac{dx}{3-x^2}$ $\quad \int \frac{dx}{3-x} = \frac{1}{2\sqrt{3}} \ln\left|\frac{\sqrt{3}+x}{\sqrt{3}-x}\right| + C$

4. $\int \frac{dx}{(4x)^2+3^2}$ $\quad \int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan(x/a) + C$

$\int \frac{dx}{16x^2+9}$ $\quad \cancel{\int \frac{dx}{a^2+x^2} \cdots}$

$\int \frac{\frac{1}{4} du}{u^2+3^2} = \frac{1}{4} \int \frac{du}{u^2+3^2}$

$\frac{1}{4}\left(\frac{1}{3}\arctan\left(\frac{u}{3}\right)\right) + C = \frac{1}{12}\arctan\left(\frac{4x}{3}\right) + C$ где $a = 3$

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